Abdullah Ibne Tayeb Tamur

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Portfolio | GitHub | Google Scholar | ORCID | LinkedIn

RESEARCH PROFILE

Research interests: Neuroimaging AI; 3D brain MRI analysis; multimodal medical imaging; domain-robust deep learning; interpretable medical computer vision; leakage-aware evaluation.

Computer Science and Engineering graduate with **six published or accepted conference papers (three as first author)** and two manuscripts under review. My first-author **CMA-Fusion** work integrates 3D brain MRI and clinical biomarkers for Alzheimer's disease detection; I also helped mentor **NeuroStack**, a concept-based brain-tumor MRI triage study. My experience includes attention-based visual modeling, PyTorch, interpretability, and careful experimental evaluation. For doctoral research, I hope to extend these strengths toward **generalizable neuroimaging models, lesion analysis, and quantitative MRI applications**.

EDUCATION

International Islamic University Chittagong (IIUC) *Bangladesh*
B.Sc. in Computer Science and Engineering Jan 2022 – Jun 2026
CGPA: 3.416/4.00 **GPA (last four semesters):** 3.71/4.00

RESEARCH EXPERIENCE

Elite Research Lab LLC *Remote*
Research Assistant Nov 2026 – Present
Research Intern Jun 2026 – Nov 2026
Develop explainable clinical AI and diagnostic ML pipelines under NDA, including retrieval-augmented and LLM-assisted systems with emphasis on interpretability, privacy, reliability, and human review.

Computational Intelligence and Society (CIS) Lab, IIUC *Chattogram, Bangladesh*
Research Mentor Jan 2024 – Present
Mentor undergraduate researchers in literature review, experimental design, model implementation, validation, and manuscript preparation. Provided research and evaluation guidance to *NeuroStack* (brain-tumor MRI triage; ICISET 2026) and *DiaXAI-Stack* (ICCA 2026), both accepted for presentation.

PUBLICATIONS – PEER-REVIEWED / ACCEPTED

[1] **Tamur, A. I. T.** et al. (**First Author**). *Cross Modality Attention for Leakage-Free Alzheimer's Disease Detection Using 3D MRI and Clinical Biomarkers.* **ICCA 2026**, accepted for presentation. Cross-modality attention for brain MRI and structured biomarkers, with leakage-aware evaluation.

[2] Islam, M., **Tamur, A. I. T.**, et al. (**Second Author; Research Mentor**). *NeuroStack: A Hybrid Concept Bottleneck Stacking Ensemble for Interpretable Brain Tumor Triage on BraTS 2024.* **ICISET 2026**, accepted for presentation. Interpretable MRI triage with concept bottlenecks, stacking, SHAP, and Grad-CAM; mentored experimental design and evaluation.

[3] **Tamur, A. I. T.** et al. (**First Author**). *CHD-Net: An Explainable Deep Learning Framework for Pediatric Congenital Heart Disease Classification.* **IEEE QPAIN 2026; Best Presenter Award**. Echocardiography-based classification with ensemble learning and SHAP. doi:10.1109/QPAIN69676.2026.11545695

[4] **Tamur, A. I. T.** et al. (Third Author). *FUSE SENet: Fusion with SE Attention Network for Robust Skin Lesion Classification.* **IEEE ICCIT 2025**. Attention-based fusion for dermatological image classification. doi:10.1109/ICCIT68739.2025.11491158

[5] **Tamur, A. I. T.** et al. (**First Author**). *Dual-Stream FreqViT: FFT and Vision Transformer Fusion for Robust Plant Disease Detection.* **IEEE QPAIN 2026**. CNN, frequency-domain features, and Transformer fusion for visual recognition. doi:10.1109/QPAIN69676.2026.11546204

[6] **Tamur, A. I. T.** et al. (Fifth Author; Research Mentor). *DiaXAI-Stack: An Explainable AI Framework for Diabetes Prediction Using Leakage-Free Stacking Ensemble Learning with Dual Threshold Optimization.* **ICCA 2026**, accepted for presentation. Mentored preprocessing, experimental design, and SHAP/LIME interpretation.

MANUSCRIPTS UNDER REVIEW

[7] **Tamur, A. I. T.** et al. (**First Author**). *HemX: An Explainable Hierarchical Machine-Learning Framework for CBC-Based Hematological Triage. Informatics in Medicine Unlocked* (Elsevier), **under review**. Hierarchical CBC-based hematological triage on **13,823 patient records**, with SHAP/LIME explanations and stratified validation.

[8] **Tamur, A. I. T.** et al. (Second Author). *Fed-NeuroTriage: Federated Distillation Architecture for Psychiatric NLP*. Under review at *JSCDM*. Privacy-aware federated learning and knowledge distillation with MIMIC-IV clinical discharge summaries.

SELECTED PROJECTS & DEPLOYMENTS

HemoForge – Explainable Hematology Platform. Research/engineering prototype for nine-class blood-count prediction with SHAP explanations, out-of-distribution checks, selective-review indicators, and a **10-node LangGraph workflow** with LangSmith tracing. Includes human-verified report extraction and a case archive; doctor/patient-facing LLM assistance is in development.

HemaTriage. CPU-oriented hematological triage web application based on HemX, designed with resource-constrained deployment settings in mind.

SpectraLeaf Analytics (Portfolio). Deployed visual disease-classification system using CNN-based spatial features and FFT-based spectral features to address field-lighting variation.

Wolf Scholar (GitHub). Multimodal retrieval-augmented research assistant built with LangChain, FAISS, and Groq for evidence-grounded multi-document analysis.

Customer Support Agent (GitHub). LangGraph application with specialized tools, guardrails, human approval, memory, and workflow observability.

TECHNICAL SKILLS & RESEARCH METHODS

Medical AI and visual learning: Medical image classification, 3D MRI and clinical-feature fusion, CNN/Transformer architectures, attention mechanisms, SHAP, LIME, Grad-CAM (research mentoring), and OOD awareness.

Evaluation methods: Leakage-aware validation, subject-level and stratified evaluation, multimodal model comparison, interpretability, and reproducible research workflows.

Languages and tools: Python, C++, SQL, PyTorch, TensorFlow, Keras, OpenCV, Git, LaTeX, Jupyter Notebook; LangChain, LangGraph, FAISS, FastAPI, Flask.

AWARDS, TEACHING & SERVICE

Best Presenter Award, IEEE QPAIN 2026, for CHD-Net; **AgentX AI Prompting Competition**, ranked 6th among 700+ IIUC participants.

Home Tutor (2020–present): Biology, English, ICT, and Chemistry tutoring. **IIUC Computer Club:** Assistant Social Welfare Secretary; peer academic support and mental-health awareness. **Flood Relief Volunteer** (2024): Collaborated with local relief groups and Bangladesh Red Crescent Society.